

Advanced Python Programming

Loop Tracing Practice Questions (No Answers)

Instructions: Predict the output of each program. Clearly show the trace of program execution and justify your answer.

Question 1

```
for i in range(1,6):
    if i%2: continue
    for j in range(i):
        if j==i//2: break
        print(j,end="")
    print("#",end=" ")
```

Question 2

```
for i in range(4):
    for j in range(4):
        if i==j: continue
        if i+j==4: break
        print(i,j,end=" | ")
```

Question 3

```
for i in range(1,5):
    for j in range(1,5):
        if i==j: break
        if (i+j)%2==0: continue
        print(i*j,end=" ")
```

Question 4

```
for i in range(5):
    for j in range(5):
        if j>i: break
        if (i*j)%2: continue
        print(i+j,end=" ")
```

Question 5

```
x=0
for i in range(1,6):
```

```
x+=i
if x%3==0: continue
print(x,end=" ")
```

Question 6

```
x=1
for i in range(4):
    for j in range(2):
        x+=i+j
    print(x,end=" ")
```

Question 7

```
for i in range(1,5):
    for j in range(4,0,-1):
        if i+j==5: break
        print(i*j,end=" ")
```

Question 8

```
x=0
for i in range(4):
    x^=i
    if x&1:
        print(x,end=" ")
    else:
        print(i,end=" ")
```

Question 9

```
for i in range(3):
    for j in range(3):
        if i==2 and j==1: continue
        print(i+j,end=" ")
```

Question 10

```
x=2
for i in range(1,5):
    x*=i
    if x%4==0:
        print(i,end=" ")
```

```
else:  
    print(x,end=" ")
```